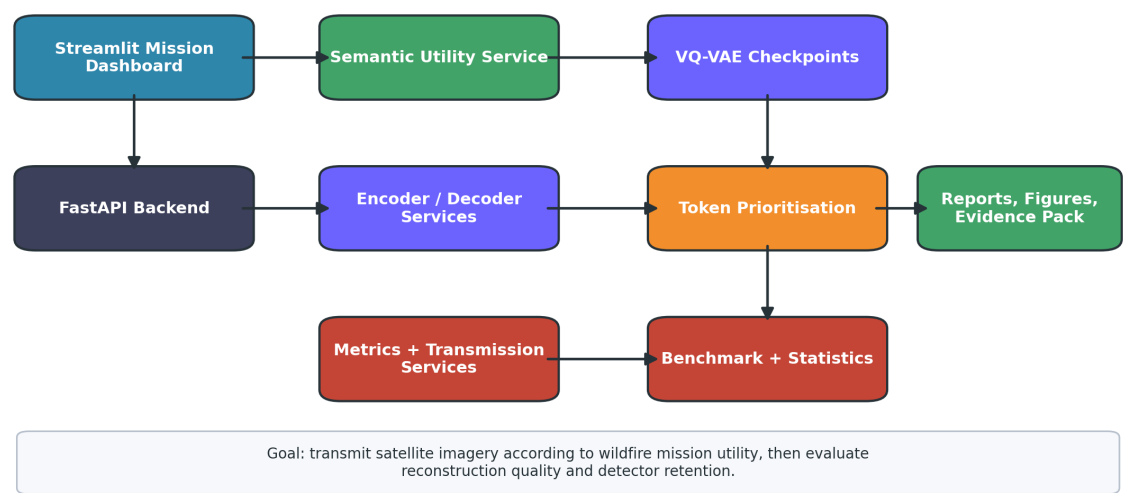


PhD Supervisor Package

Semantic Utility-Aware Compression for Wildfire-Centric Earth Observation Systems

CompressAI System Architecture



Purpose: a supervisor-ready research package with problem framing, methodology, architecture, experiments, results, limitations, and a 4-year PhD plan.

1. Executive Summary

This project investigates semantic communication for Earth Observation satellites. The central idea is to transmit learned image tokens according to wildfire mission utility instead of compressing every pixel equally. The current system uses a VQ-VAE encoder/decoder, semantic utility estimation, utility-aware token ranking, satellite communication metrics, and detector-retention evaluation.

The key scientific finding so far is that better visual reconstruction does not automatically mean better mission utility. In the 500-patch Sentinel-2 benchmark, the regularized mixed-domain VQ-VAE improves PSNR and SSIM, but it reduces detector retention. This motivates a PhD research programme on utility-preserving representation learning and mission-aware semantic compression.

Core research hypothesis

Semantic utility-aware token prioritisation preserves wildfire-relevant Earth Observation information more effectively than conventional distortion-optimised compression under equivalent bandwidth constraints.

Supervisor relevance

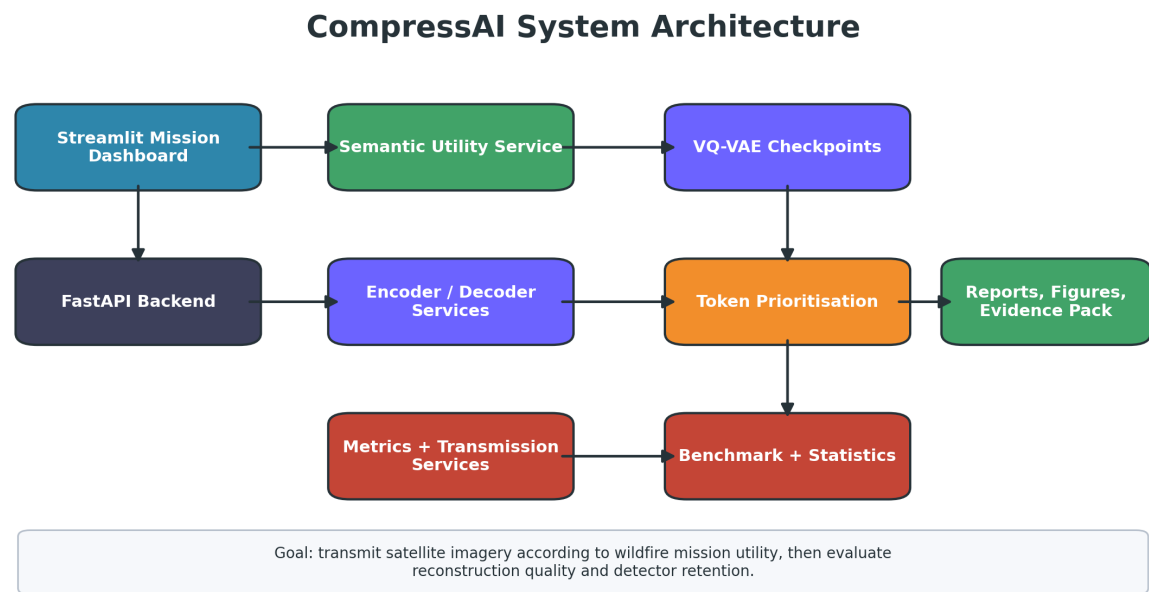
- Earth Observation and remote sensing: Sentinel-2 wildfire-relevant validation and satellite imagery constraints.
- Onboard AI: token selection under constrained communication budgets.
- Semantic communication: transmission of task-relevant information rather than all pixels.
- Neural compression: VQ-VAE tokenisation, reconstruction, and detector-retention trade-offs.

2. Research Gap and Questions

Most image compression research optimises distortion, perceptual quality, or bitrate. Earth Observation systems often evaluate downstream interpretation separately from compression. This project addresses the gap between compression quality and mission value by directly measuring whether wildfire-relevant information survives compression and reconstruction.

Research question	Methodology	Measurable outcome
RQ1: Can wildfire relevance be represented as a semantic utility signal?	Utility maps from wildfire, smoke, burn-scar, saliency, and detector cues.	Utility-map coverage, relevance retention, region preservation.
RQ2: Can VQ-VAE tokens be prioritised by mission utility?	Token-grid mapping, utility-aware ranking, token pruning.	SUS, detector retention, token count, compression ratio.
RQ3: Does mission utility diverge from visual quality?	Compare original and fine-tuned VQ-VAE checkpoints.	PSNR/SSIM versus SUS/detector-retention trade-off.
RQ4: Is this practical for satellite communication?	Payload estimation and bandwidth-saving analysis.	Bandwidth saved, compression ratio, transmission reduction.

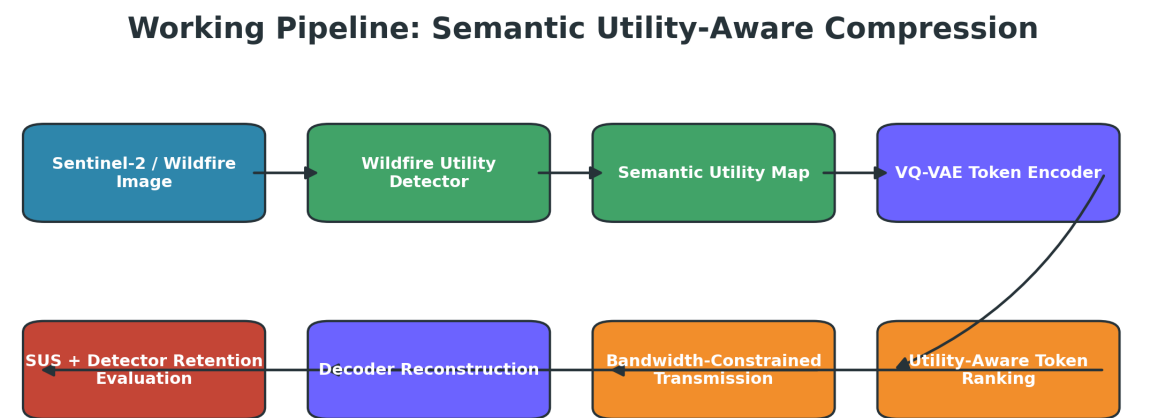
3. System Architecture



The implementation is modular: the frontend supports mission-control-style interaction, FastAPI exposes compression and analysis endpoints, service layers isolate semantic analysis and VQ-VAE encoding/decoding, and the benchmark layer produces CSV, JSON, figures, and statistical reports.

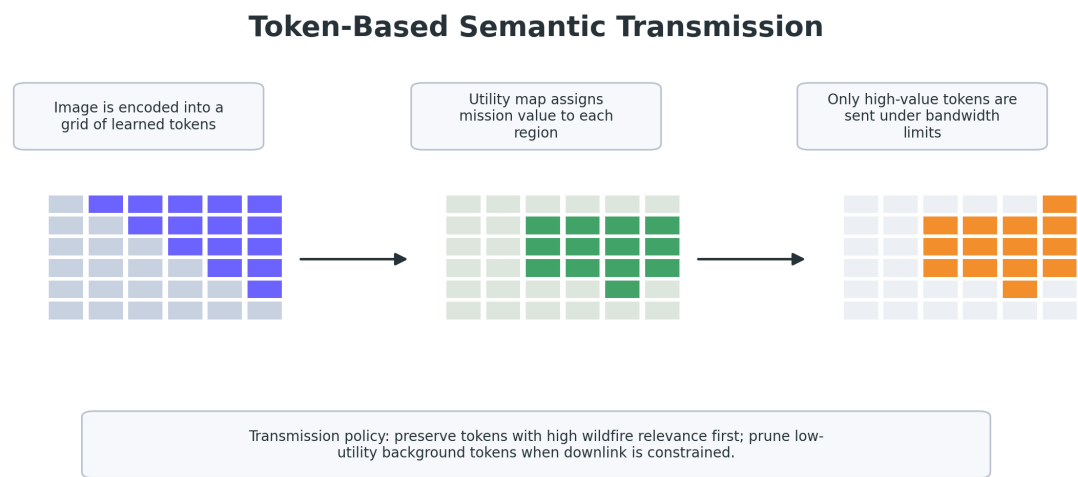
Module	Role in research platform
Semantic Utility Service	Detects or estimates wildfire-relevant regions and produces mission utility maps.
Encoder / Decoder Services	Load VQ-VAE checkpoints and perform tokenisation and reconstruction.
Token Prioritisation	Ranks tokens using mission utility, entropy, and communication cost.
Metrics and Transmission Services	Compute SUS, detector retention, PSNR, SSIM, compression ratio, and bandwidth savings.
Benchmark and Statistics	Runs paired experiments, bootstrap confidence intervals, t-tests, Wilcoxon tests, and result tables.

4. Working Methodology



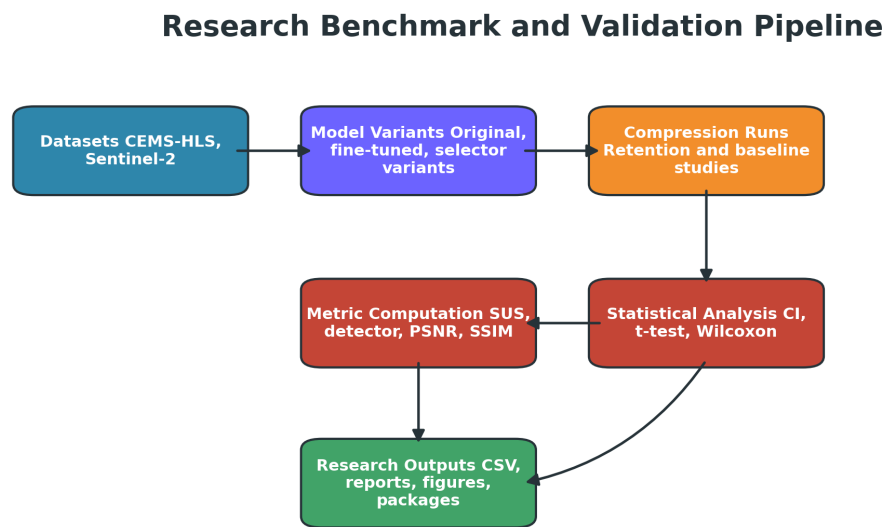
The pipeline receives a Sentinel-2 or wildfire image, estimates semantic utility, encodes the image into learned tokens, ranks tokens according to mission value, simulates bandwidth-constrained transmission, reconstructs the image, and evaluates semantic utility retention.

5. Token Transmission Mechanism



The system treats the encoded image as a token grid. Important wildfire-relevant regions receive higher priority, while low-utility background regions can be pruned more aggressively. This creates an experimental semantic communication system rather than a standard distortion-only compression pipeline.

6. Benchmark and Validation Workflow

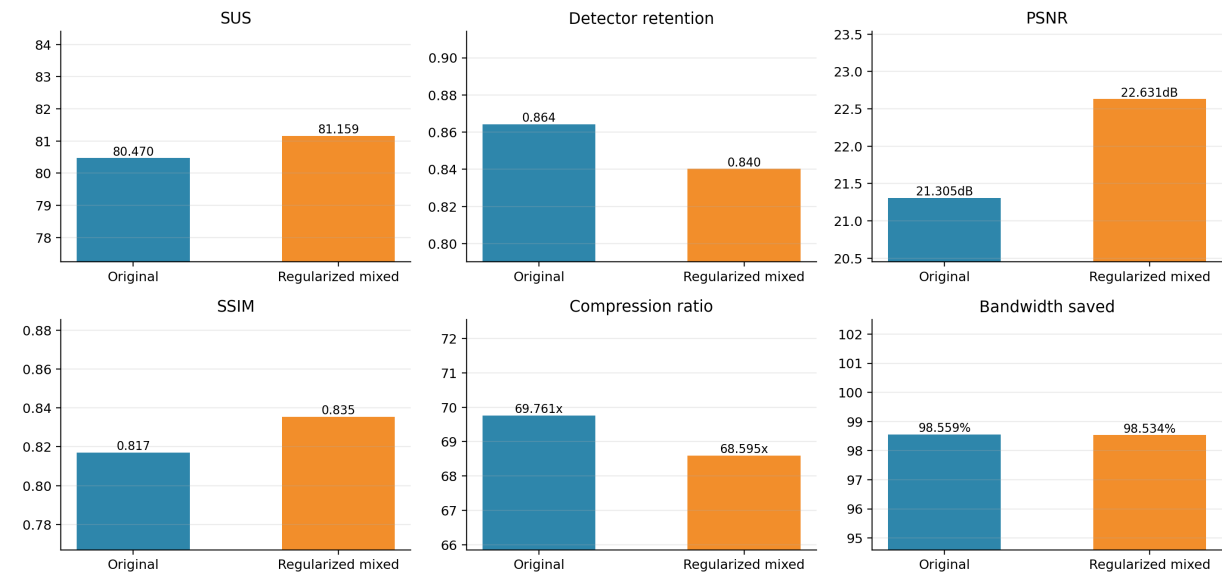


Benchmark element	Current implementation
Datasets	CEMS-HLS wildfire imagery and Sentinel-2 benchmark patches.
Model variants	Original VQ-VAE, CEMS-only fine-tune, mixed-domain fine-tune, regularized mixed-domain fine-tune.
Metrics	SUS, detector retention, PSNR, SSIM, compression ratio, bandwidth saved, inference latency.
Statistics	Paired t-test, Wilcoxon signed-rank, bootstrap confidence intervals.
Decision rule	Do not promote a checkpoint if detector retention drops significantly, even if PSNR/SSIM improve.

7. Sentinel-2 500-Patch Evidence

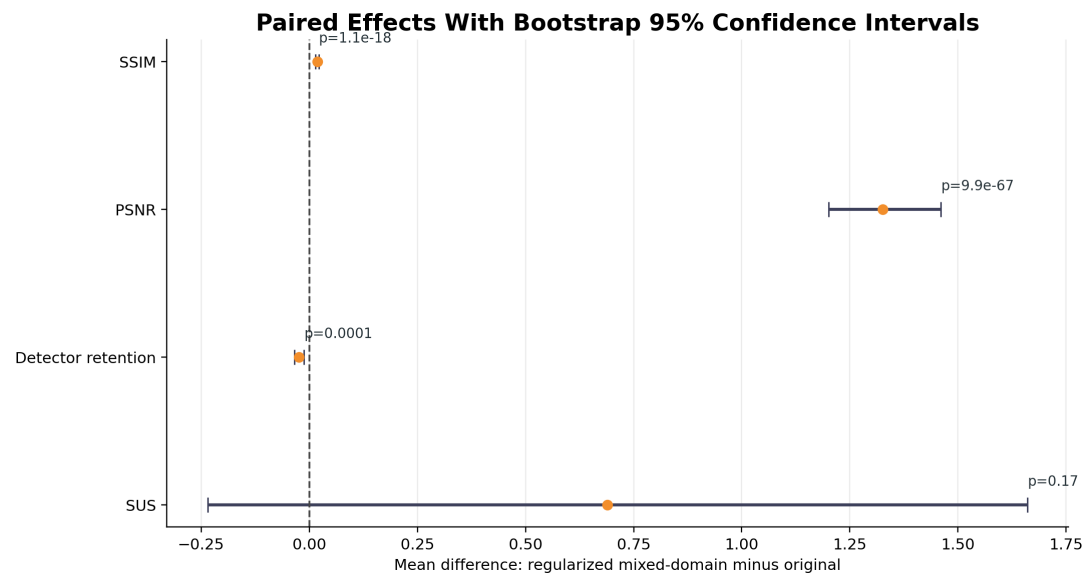
Metric	Original VQ-VAE	Regularized mixed-domain VQ-VAE	Interpretation
SUS	80.470	81.159	Candidate slightly higher, but not significant enough alone for promotion.
Detector retention	0.8642	0.8403	Original is safer for wildfire mission-utility preservation.
PSNR	21.305	22.631	Candidate improves visual reconstruction by +1.326 dB.
SSIM	0.8170	0.8354	Candidate improves structural similarity.
Compression ratio	69.76x	68.60x	Original remains slightly stronger for compression ratio.
Bandwidth saved	98.56%	98.53%	Both preserve very high communication savings.

Sentinel-2 500-Patch Benchmark: Original vs Regularized VQ-VAE



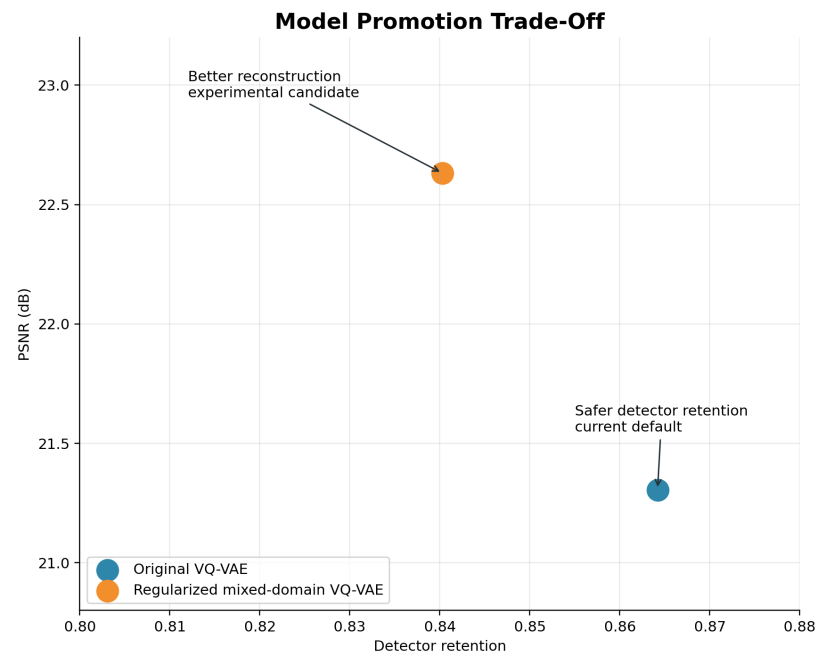
This figure is the most important current result for supervisors. It shows why the project is not merely an engineering demo: mission utility and conventional image quality disagree. The candidate model looks better under PSNR and SSIM, but the original model is safer for detector retention.

8. Statistical Interpretation



Metric	Mean difference	Paired t-test p	Bootstrap 95% CI	Decision
SUS	+0.6895	0.1741	[-0.2347, 1.6611]	Not conclusive by paired t-test.
Detector retention	-0.0239	0.000105	[-0.0347, -0.0120]	Significant drop; do not promote candidate as default.
PSNR	+1.3261 dB	9.91e-67	[1.2015, 1.4605]	Strong reconstruction gain.
SSIM	+0.0184	1.15e-18	[0.0144, 0.0220]	Strong structural gain.

9. Model Promotion Decision



Decision: the regularized mixed-domain checkpoint should not replace the original checkpoint yet. It is a strong reconstruction-improved candidate, but the detector-retention drop is statistically significant. This makes the original checkpoint the safer public default for mission-utility experiments.

10. Contributions and Publication Potential

Contribution	Why it is publishable
Semantic Utility Score and detector-retention evaluation	Moves evaluation beyond PSNR/SSIM toward mission-specific EO usefulness.
Utility-aware token prioritisation	Demonstrates semantic communication using learned image-token representations.
Satellite communication framing	Connects compression to bandwidth, downlink, and resource-constrained operation.
Cross-model checkpoint evidence	Shows that satellite fine-tuning can improve visual quality while harming mission utility.
Reproducible platform	Supports further ablation, benchmark, and onboard AI experiments.

11. Limitations

- Current detector and utility maps are research-grade and require further validation against expert wildfire labels.
- The regularized VQ-VAE is not yet the default because detector retention drops on the larger Sentinel-2 test.
- JPEG/JPEG2000 remain strong visual baselines and should be positioned honestly.
- More geographically diverse Sentinel-2 wildfire scenes are needed for journal-level claims.
- Onboard deployment has been simulated; hardware validation on Jetson or satellite-class compute remains future work.

12. Four-Year PhD Plan

Year	Research focus	Expected outputs
1	Formalise wildfire utility metrics, strengthen datasets, validate semantic utility maps.	Workshop paper, reproducible benchmark, supervisor-aligned proposal.
2	Develop utility-preserving token selection and detector-retention-aware representation learning.	Conference paper, open-source model comparison.
3	Evaluate realistic satellite communication constraints and cross-dataset Earth Observation generalisation.	Journal paper, mission-scale bandwidth analysis.
4	Optimise for onboard deployment and finalise thesis contributions.	Thesis, final journal paper, documented platform release.

Best use: send this PDF with a short email, GitHub link, and a 2-page summary. It is designed to help a supervisor understand the problem, the evidence, and the PhD direction quickly.